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Dated: _____

Sec: 9

: AI-BASED EMERGENCY RESPONSE AGENT:

1. Identify The Problem In The Context Of An Agent

Problem Statement: In universities and public environments emergencies such as fires, medical incidents, accidents or security threats may occur unexpectedly. Delayed detection and response can lead to serious consequences, including injuries, property damage and safety risks.

Agent Goal: The AI-based Emergency Response Agent aims to detect emergency situations, assess their severity and provide immediate alerts to the appropriate authorities for a rapid response.

2. Problem Formulation:

Initial State: The environment is operating normally, and no emergency has been detected.

Action:

- Collect data from sensors and monitoring systems.
- Detect potential emergency situations.
- ^{Assess} Analyze the level of risk.
- Analyze & classify the emergency type.
- Generate alerts and notifications.
- Guide emergency response teams.

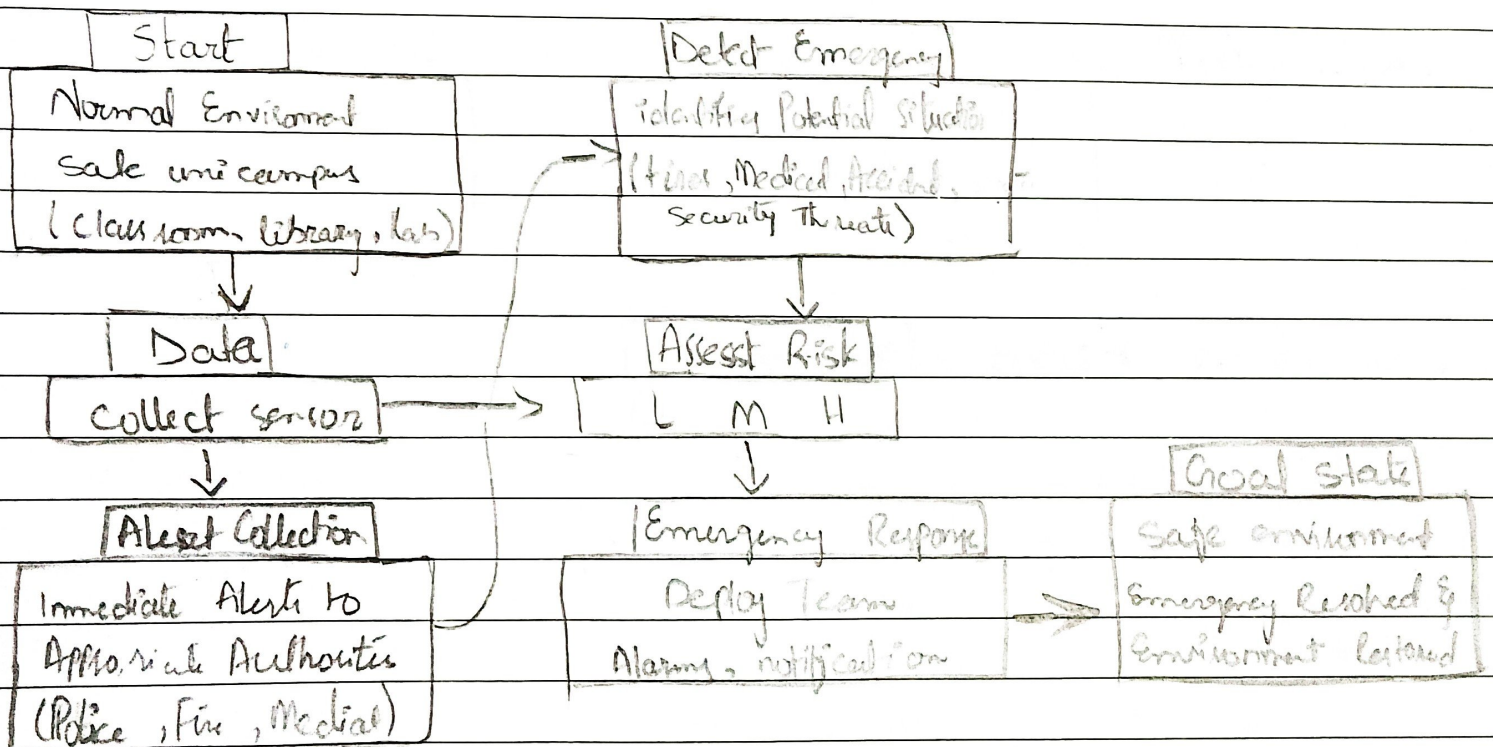
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Goal State: The emergency is detected quickly, appropriate actions are taken and the environment is restored to a safe condition.

Path cost: Minimize:

- Response time
- Damage to property
- Risk to human life

Graphical Representation:



3. PEAS Analysis:

Performance Measure: High detection accuracy, fast response time, minimum damage.

Environment: University campus, classrooms, laboratories, staff, students.

Dated: _____

Accurators: Alarms, emergency alert, notification.

Sensors: Cameras, smoke detectors, temperature sensor, emergency button.

4 Environment Properties of The Agent:

Property	Type
Observability	Partially Observed (Incomplete environment data.)
Deterministic / Non-Deterministic	Non-Deterministic (Unpredictable event outcomes)
Static / Dynamic	Dynamic (Environment constantly changes.)
Discrete / Continuous	Continuous (Real-time sensor streaming)
Single-Agent / Multi-Agent	Multi-Agent. (Interacts with external system)